The Association Between the Kaitz Index and Employment in Puerto Rico (Working Paper)

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Abstract

This study examines the relationship between the Kaitz Index and employment levels in Puerto Rico using panel data at the postal code level. The data covers the period from Q1 2012 to Q4 2023. A Spatial Durbin Model with fixed effects is applied to investigate both the direct effect of the Kaitz Index on employment and potential spatial effects from adjacent regions. The findings largely support existing literature suggesting a negative relationship between the minimum wage and employment. Additionally, despite some convergence issues with the MCMC of the Kaitz variable, the confidence interval indicates a negative spillover effect on neighboring regions' employment levels.

1 Introduction

The minimum wage and its impact on employment are among the most debated topics in labor economics, especially in regions with unique economic structures like Puerto Rico. Numerous studies have attempted to analyze this relationship, yielding diverse results. For example, a meta-analysis by Stanley (2009) [6] and Card & Krueger (1994) [3] concluded that there is no significant effect on employment, while the Federal Reserve's 2012 *Competitiveness of the Puerto Rican Economy* (2012) [9] report emphasized the need for policies that focus on employment creation and suggested a subminimum wage for young workers under 25. Locally, Hernández (2017) found a reduction in total employment following minimum wage increases, while Hernández & Wu (2025) identified differential responses by local versus foreign-owned businesses. However, the spatial dynamics of the minimum wage and how wage conditions in one region may affect neighboring areas—remain under explored. This paper seeks to fill that gap by analyzing the relationship between the Kaitz Index [10] and employment using spatial econometrics.

2 Literature Review

Castillo (1983) [4] notes that Puerto Rican emigrants to the U.S. often migrate due to unemployment, a condition linked to minimum wage policies. Freeman (1992) [5] found that the imposition of the U.S. minimum wage in Puerto Rico led to an 8%–10% reduction in employment. Brown (1988) [1] noted that the U.S. minimum wage altered the earnings distribution in Puerto Rico, creating sharp peaks. Krueger (1994) [3] found a negative relationship between minimum wages and employment using aggregate time series data. In contrast, Caraballo-Cueto (2016) [2] observed a slight positive effect of minimum wages on employment in the short run. Finally, Neumark (2000) [8] found that higher minimum wages are associated with higher unemployment rates across U.S. regions. These studies highlight the complexity of the minimum wage-employment relationship, with results varying by geography, time and methodology.

3 Methodology

3.1 Calculating the Kaitz-index

The minimum wage data used in this study comes from the FRED database, while employment and average wage data are sourced from the Quarterly Census of Employment and Wages (QCEW) provided by the Department of Labor. The dataset consists of 6,240 observations spanning Q1 2012 to Q4 2023, organized by postal codes in Puerto Rico. Additional data on economic characteristics of the zip codes were obtained from the American Community Survey. The following table shows how the Kaitz index is defined for this study.

$$K_{it} = \frac{m_t}{\bar{w}_{it}}$$

where:

- K_{it} is the Kaitz Index in zip code i at time t,
- m_t is the minimum wage at time t,
- $\bar{w}_{it}$ is the average wage in zip code i during time t.

3.2 Introducing Space

This study employs a Spatial Durbin Model (SDM) [7] with fixed effects to account for spatial dependence between regions. The model allows for the estimation of both direct and indirect effects. The Kaitz Index is defined as:

The full Spatial Durbin Model is:

$$Y_{it} = \alpha + \beta_1 X_{it} + \rho W Y_{it} + \gamma W X_{it} + \epsilon_{it}$$

where:

- Y_{it} is employment in zip code i at time t,

- X_{it} includes the Kaitz Index (K), the minimum wage (M), and the average wage (W),
- W is the spatial weights matrix reflecting adjacency between regions,
- ρ is the spatial autoregressive parameter capturing spatial dependence,
- ϵ_{it} is the error term.

For robustness this research implements 4 distinct W spatial weights. They are defined as followed

- W_{queen}: Which models all regions that connect borders

$$Y_{it} = \alpha + \beta_1 X_{it} + \rho W_{queen} Y_{it} + \gamma W_{queen} X_{it} + \epsilon_{it}$$

- W_{knn6}: Which models all regions that up too 6 neighboring regions away

$$Y_{it} = \alpha + \beta_1 X_{it} + \rho W_{knn6} Y_{it} + \gamma W_{knn6} X_{it} + \epsilon_{it}$$

- $W_{distance}$: Which model all regions in a 30 mile radius.

$$Y_{it} = \alpha + \beta_1 X_{it} + \rho W_{distance} Y_{it} + \gamma W_{distance} X_{it} + \epsilon_{it}$$

- W_{tensor}: A semi parametric approach is implemented as a diagnostic.

$$Y_{it} = \alpha + \beta_1 X_{it} + f(C_i) + \epsilon_{it}$$

The W_{tensor} approach is used to determine if the assumption that spatial correlation is uniform between regions. Where $f()$ is the tensor product using the C_i which is the centroid for the region.

4 Results

The results indicate a negative and statistically significant relationship between the Kaitz Index and employment in Puerto Rico. Additionally, the spillover effects of the Kaitz Index on neighboring regions' employment levels are negative. However, convergence issues with the MCMC method for the Kaitz variable make it difficult to quantify the precise magnitude of the spillover effects. A small but significant effect of households with children under 6 years old on employment was also observed. This convergence issues might stem from the possibility of miss specified weights matrix. Looking at the spatial smoothing, they are not uniform this hints to the possibility that the spatial relationships from the regions are not uniform between each other

5 Conclusion

This study found a negative and statistically significant relationship between the Kaitz Index and employment in Puerto Rico's ZIP codes, suggesting that higher minimum wages relative to the average wage may reduce employment levels. The spatial econometric approach used here also highlights spillover effects, indicating that wage policies in one area can influence neighboring regions. Future research should explore lagged spatial effects to evaluate whether the historical characteristics of neighboring regions significantly influence current employment trends. Additionally there other more complex methods like spatial smoothing should be consider given that their spatila relationships do not seem to be consistent with each other.

References

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Appendix A: Supplementary Tables

Variables	**Mean**	**Standard Deviation**
K Index	0.72	0.21
ZIP Employment	26.93	25.77
Commute by Car	6543.96	5363.64
Own Children under 6	1586.17	1196.55
Own Children 6-17	3920.28	2918.55
Food Stamp Participation	3727.97	2447.43
With Social Security	42.06	43.28

Table A1: Descriptive Statistics for Main Variables

Variable	**Mean**	**3% CI (Lower)**	**3% CI (Upper)**
Kaitz Index	-9.594	-12.806	-6.093
Sigma	8.164	8.013	8.300
Intercept	13.808	3.715	23.321
Commute by Car	-0.001	-0.001	-0.000
Own Children under 6	0.005	0.004	0.007
Own Children 6-17	0.000	-0.001	0.001
Food Stamp Participation	-0.001	-0.002	-0.000
Social Security Disability	0.003	-0.007	0.013
WQ Employment	-0.040	-0.106	0.036
WQ K Index	-1.139	-8.625	7.089

Table A2: Regression Results: Parameter Estimates and Confidence Intervals (Queen)

Variable	**R-Hat**
Kaitz Index	1.00
Sigma	1.00
Intercept	1.01
Commute by Car	1.01
Own Children under 6	1.00
Own Children 6-17	1.00
Food Stamp Participation	1.00
Social Security Disability	1.00
WQ Employment	1.00
WQ K Index	1.00

Table A3: R-Hat for Individual Variables (queen)

Variable	Mean	3% CI (Lower)	3% CI (Upper)
Kaitz Index	-9.687	-12.917	-6.293
Sigma	8.163	8.015	8.307
Intercept	14.758	4.748	25.399
Commute by Car	-0.001	-0.001	-0.000
Own Children under 6	0.005	0.004	0.007
Own Children 6-17	0.000	-0.001	0.001
Food Stamp Participation	-0.001	-0.002	-0.000
Social Security Disability	0.002	-0.008	0.012
WK6 Employment	0.044	-0.031	0.113
WK6 K Index	-2.916	-11.209	5.525

Table A4: Regression Results: Parameter Estimates and Confidence Intervals (KNN6)

Variable	R-Hat
Kaitz Index	1.02
Sigma	1.00
Intercept	1.11
Commute by Car	1.02
Own Children under 6	1.01
Own Children 6-17	1.01
Food Stamp Participation	1.03
Social Security Disability	1.01
WK6 Employment	1.02
WK6 K Index	1.08

Table A5: R-Squared for Individual Variables (KNN6)

Variable	Mean	3% CI (Lower)	3% CI (Upper)
Kaitz Index	-9.650	-13.154	-6.244
Sigma	8.160	8.009	8.307
Intercept	17.368	-5.015	45.195
Commute by Car	-0.001	-0.001	-0.000
Own Children under 6	0.005	0.004	0.007
Own Children 6-17	0.000	-0.001	0.001
Food Stamp Participation	-0.001	-0.002	-0.000
Social Security Disability	0.004	-0.005	0.014
WD30 Employment	-0.301	-0.519	-0.090
WD30 K Index	-0.084	-28.740	23.926

Table A6: Regression Results: Parameter Estimates and Confidence Intervals (Distance)

Variable	R-Hat
Kaitz Index	1.03
Sigma	1.00
Intercept	1.03
Commute by Car	1.02
Own Children under 6	1.00
Own Children 6-17	1.00
Food Stamp Participation	1.01
Social Security Disability	1.00
WD30 Employment	1.01
WD30 K Index	1.04

Table A7: R-Squared for Individual Variables (Distance)

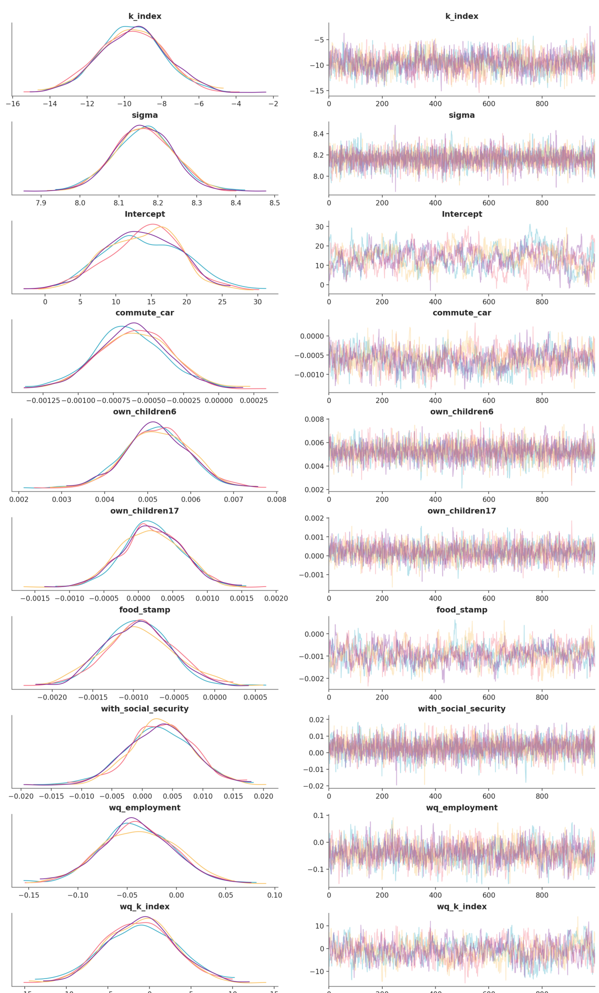

Figure 1: Summary of Regression Results (Queen)

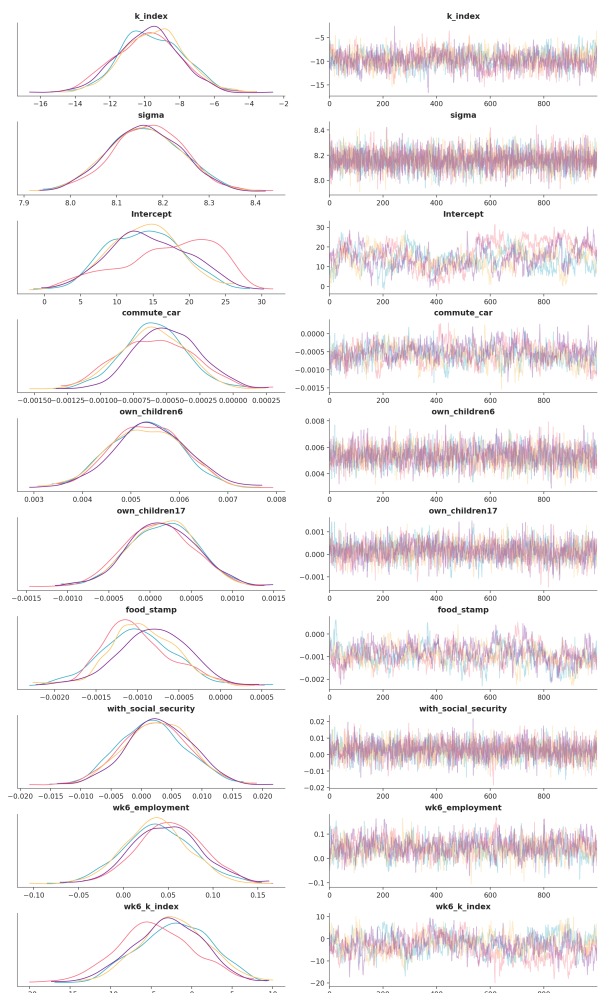

Figure 2: Summary of Regression Results (knn6)

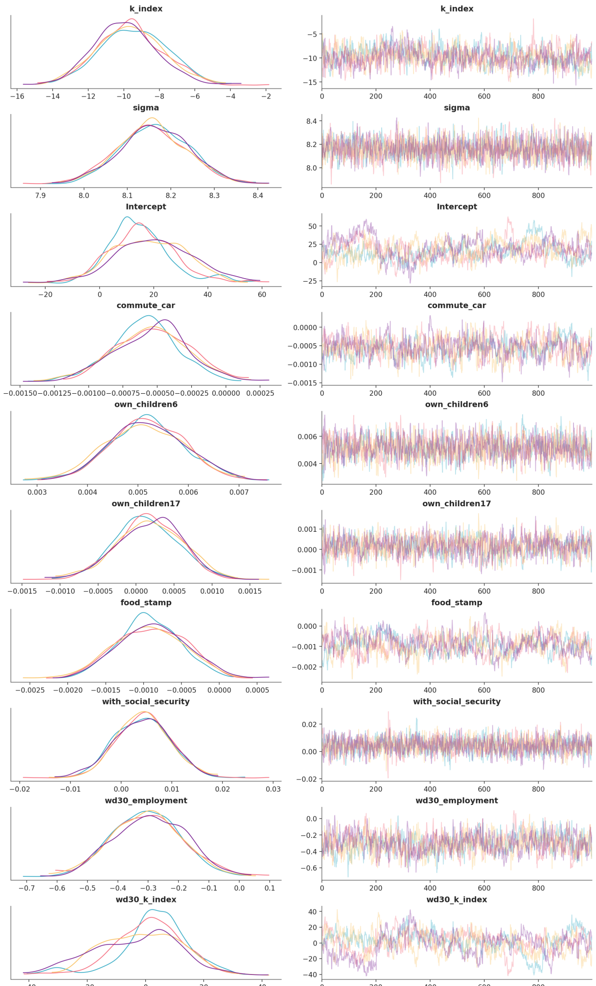

Figure 3: Summary of Regression Results (Distance)

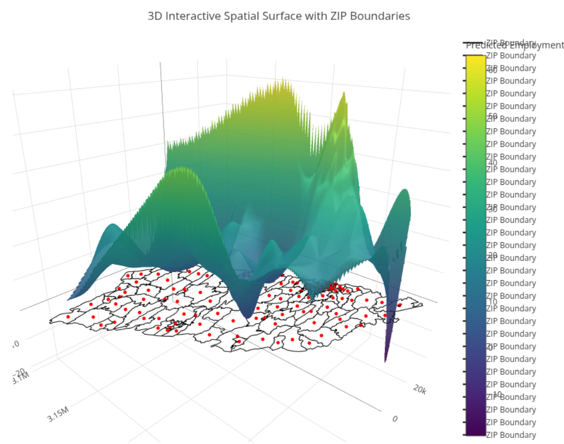

Figure 4: Tensor Diagnostic